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Facial Expression Recognition Using Multi-Branch Attention Convolutional Neural Network

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ABSTRACT Facial expression recognition technology was extensively used. This paper develops a multi-branch attention convolutional neural network based on a multiple-branch structure to recognize facial expressions. First, features are extracted from facial images in a multi-branch architecture, and features from three branches are fused. Second, to address the issue of insufficient feature extraction and poor recognition performance, the Convolutional Block Attention Module is added as attention module. Third, our model reduces parameters and computation loads by using depth-wise separable convolutions. The experiments on the FER2013, FERPLUS, and CK+ datasets show that the recognition rates of the proposed model are 69.49%, 84.633%, and 99.39%, respectively. The proposed method has a higher efficiency in extracting image features than traditional deep learning counterparts and achieves high accuracy without complicated artificial feature technology.

INDEX TERMS Facial expression recognition, multi-branch architecture, depth-wise separable convolutions, Convolutional Block Attention Module

I. INTRODUCTION

Human facial expressions can convey rich emotional information. In interpersonal communication, people can infer the emotional states of other people through facial expressions. Thus, facial expression recognition (FER) technique is extensively used for applications such as the intelligent security, the treatment of depression, customer relationship management, human-computer interaction. [1]. Automatically recognizing facial expression can enable more effective communicate and thus provide better accurate services in above mentioned applications. Consequently, FER has become a hot research topic in a pursuit for a full realization of its scientific and practical potentials [2].

With the great development of deep learning technologies, many research works via deep learning have achieved remarkable achievement. As a popular deep learning network model, convolutional neural network (CNN) has been extensively adopted for image detection and classification applications. A well-designed CNN model can extract generic features of a low level and semantic features of a high level simultaneously by training on millions of images, and improve the performance of recognition [1]. For example, AlexNet [3] proposed by Krizhevsky et al. is a classical CNN and won the championship in the ImageNet Challenge 2012. Since then, there have been many networks proposed, like VGG [4], ResNet [5], Inception [6], etc. Hence, we designed our algorithm based on CNN to achieve better FER performances.

Most image recognition algorithms based on deep learning include two parts: feature extraction and classifier design. How to extract feature maps is core research. Although the CNN base model has been considered efficient in extracting image features, however, it cannot effectively exploit the importance and correlation of feature maps. So, attention mechanisms [7] are proposed and designed to achieve selective focus on important information, emphasizing useful features, and reducing redundancies.

Deep learning networks make the main contribution to the largest advances in image recognition performance [8]. However, deep learning networks often suffer from overwhelming parameters, low computational efficiency, and high memory consumption. As a result, the model can hardly be applied for portable mobile devices with limited hardware resources [9]. To solve these problems, researchers proposed network pruning, knowledge distillation, parameter quantization, dynamic computation, and architecture design. In 2017, Google proposed a lightweight model called Mobilenet V1 [10]. The Mobilenet has smaller parameters and less calculation amount by an ingenious design, and is suitable to be deployed on mobile devices. Due to the fact that most of FER applications are running on mobile devices or embedded equipment, it is necessary to design a model with acceptable parameters and computational loads.

Recently, many methods have been developed via deep learning networks for facial expression recognition. In general, deepening the number of network layers can improve nonlinear representation ability and get a better fitting effect for complex features. For example, the classical ResNet-152 is a network with 152 depth layers and a large convolutional. ResNet achieved good classification results in many classification experiments. However, for the FER application, the difficulty comes from the small data set and limited pixels of images. In this case, recognition performance is hard to be improved by increasing the number of network layers. This is because some important features from shallow layers are easily vanished in the deep layers. Therefore, this paper proposes a multi-branch method that aims to attain sufficient features from shallow layers and multiscale feature maps.

The contributions of this paper are:

1) A lightweight multi-branch convolutional neural network that combines convolutional block attention module was proposed for facial expression recognition.
2) It was found that too many layers of the network will reduce the classification accuracy, which was validated by the comparison between the proposed deep network and the other deep network models.
3) Without extending datasets or artificial features, the image augmentations technique was used to augment the training samples and achieve higher accuracy.

The organization of the rest paper is: Section 2 introduces the related work. Section 3 elaborates on the methodology of the proposed model in detail. After that, Section 4 presents the algorithm experiments, and the experimental results are evaluated and comparatively analyzed. At last, Section 5 concludes this work.

II. RELATED WORK

For FER, image preprocessing is the first step, which can effectively enhance the model's performance. Some researchers exploit to extract features by hand-crafted features with traditional methods. For example, Histogram of Oriented Gradient (HOG), Local binary pattern (LBP), Scale Invariant Feature Transform (SIFT), etc. have been demonstrated to be effective in image recognition. Yanpeng Liu et al. [11] obtain salient areas from faces, and then extract LBP and HOG features and fuse them. They applied several different classifiers to recognize six expressions, and the best recognition accuracy was 97% on CK+ datasets. Based on LBP and Adaboost, Ying Zilu et al. [12] developed a method for facial expression recognition. Although hand-crafted features can effectively improve accuracy, however, the complexity of the design for artificial features is the major drawback of these methods. In this paper, an approach free from artificial features is proposed.

As deep learning and computer technology develops rapidly, researchers have proposed many deep neural network models, including AlexNet, RCNN, VGGNet, ResNet, GoogLeNet [13], etc. These models have been used extensively and have acquired impressive results. So far, many studies have been carried out for FER by using deep learning to classify facial expressions. For example, Panagiotis Giannopoulos et al. [14] applied GoogLeNet and AlexNet to facial expressions recognition. The deep learning model is applied to feature exaction and classification. Convolutional layers are responsible for processing the input image and extracting feature maps. Both of the models are trained on FER2013 datasets. The model classifies different facial expressions with an accuracy of approximately 60%. Ruhua Lu et al. [15] conducted facial expression recognition through VGGNet, and realized an accuracy of 83.57% on CK+ datasets. GoogleNet consists of a basic structure called 'inception' which enabled the network to select receptive field information adaptively can and alleviate the problem of gradient vanishing. Ali Mollahosseini et al. [16] presented a novel deep neural network with Inception style modules for FER, and a 64.4% average accuracy was obtained. ResNet is an improved model published by He et al. in 2016. ResNet adopted residual learning to deal with the degradation of deep networks. It uses a shortcut connection with identity mappings to improve information propagation in both forward and backward passes. ResNet has been widely merged in plenty of models and achieves better performance.

Tee Connie et al. [17] presented a hybrid approach by combining CNN and SIFT to classify facial expressions. They realized over 70% classification accuracy. R. T. Ionescu et al. [18] provided a method based on the BoW (bag of words) model and realized 67.484% accuracy on the FER2013 (private test datasets).

Faced with too many parameters and heavy computation burden, several methods have been put forward to reduce network parameters and make the network more compacted. The Mobilenet deep neural network is lightweight, and its main idea is to use depthwise separable convolution layers to disassemble a standard convolution into a pointwise convolution (DSC) and a depthwise convolution. The elegant design makes MobileNet lighter and more simplified [19]. NING ZHOU et al. [20] detected real-time facial expressions by using the residual modules and depthwise separable convolution layers, and an accuracy of 67% was obtained on the FER-2013 dataset.

It is known that the region of the face contains lips, eyebrows, and nose. Features of the different regions on the face often have different contributions to facial expression classification. Zheng Lian et al. [21] split the face into seven areas and investigated the role of different facial areas in emotion recognition and conducted experiments based on a DenseNet-BC network.

In recent years, attention mechanisms have become a promising integral part of the model in image recognition,

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image segmentation, natural language processing, etc. [7]. The idea of attention mechanism originated from the human visual system. It promotes the network to automatically focusing on important features and improves the representation of core information. Xiao Sun et al. [22] propose a robust vectorized CNN with attention mechanism. They applied attention to perform region of interests relation calculation. Shervin Minaee et al. [23] presented an end-to-end attentional convolutional network that focuses on feature-rich areas in the human face, and the best recognition accuracy of 70.02% was achieved.

Insufficiency of training samples is a common challenge for FER task. Data Augmentation, Transfer Learning, Hybrid Features, etc. have been applied in many researches. Bita Houshmand et al. [24] proposed a geometric model and performed transfer learning to classify facial expressions under severe occlusion. They realized the classification accuracy of 79.98% (VGG backbone), and 79.90% (ResNet50 backbone) based on FERPLUS datasets. Ming Li et al. [25] fused ResNet and DeepID, and concatenated the feature maps for joint facial expression learning. Their model realized an accuracy of 99.31% on CK+ datasets.

Recently, some researchers are exploring multi-branches methods for computer vision tasks. Qiangchang Wang et al. [26] advanced a hierarchical pyramid diverse attention (HPDA) network to recognize faces. They used several local branches of different scales at varying hierarchical layers to learn diverse local representations. In this work, we employ multi-branch architecture for extracting multi-scale features from facial images and proposed a lightweight attention module based on CBAM. The proposed model achieved better performances with fewer parameters.

III. PROPOSED METHOD

This paper aims to develop a multi-branch attention convolutional neural network, which simultaneously predicts the multi-level labels of facial images through multiscale feature maps. The structure of proposed model is introduced in this section.

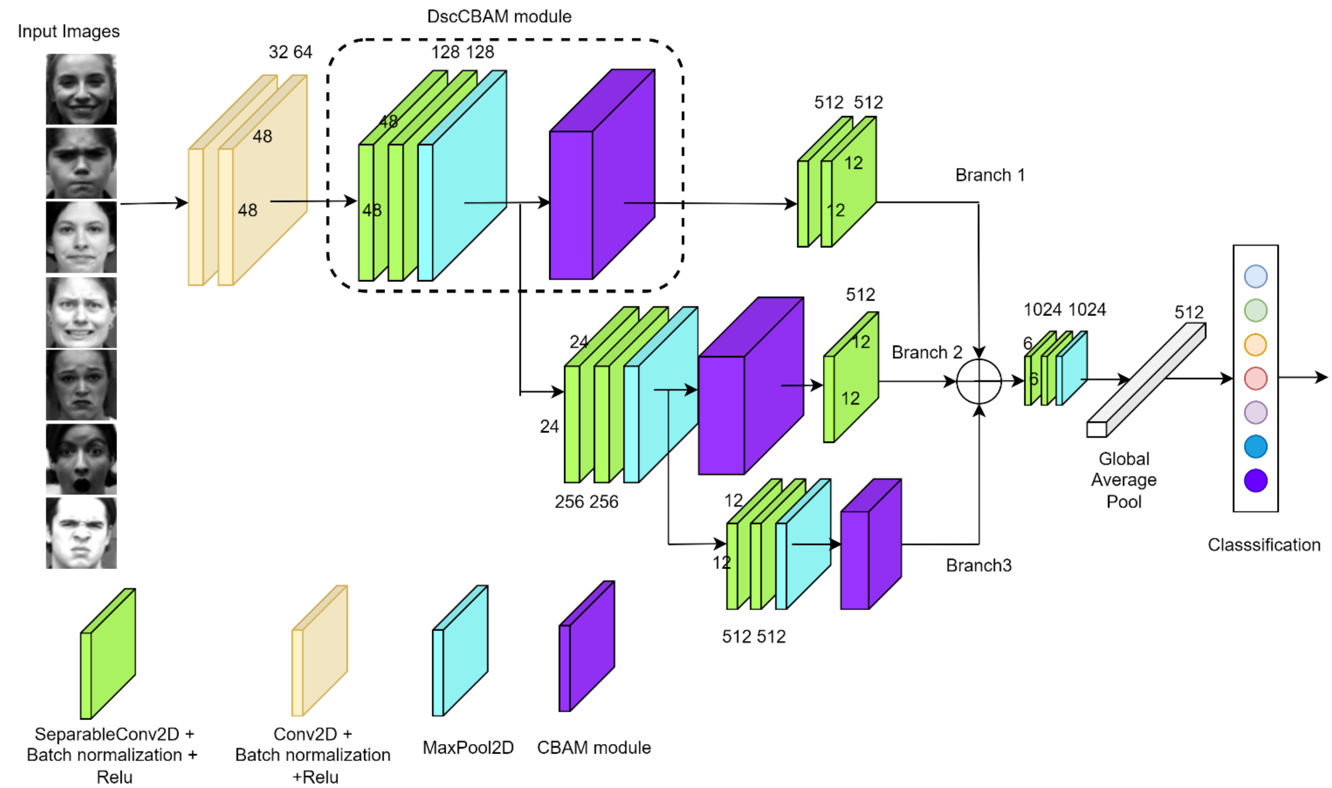

FIGURE 1. **Multi-Branch Attention Attention Convolutional Neural Network**

A. ARCHITECTURE OF MODEL

A description of the multi-branch attention CNN for FER is pictorially given in Fig. 1. The network takes a grayscale image with [48 × 48] pixels as an input to the network. The first two 3×3 convolutional layers perform a convolutional operation on patches of the input image. After the convolutional layer, there is a ReLu [27] activation function and batch normalization layers to overcome the covariate shift problem. Subsequently, the feature map passes through a multi-branch module.

In contrast to traditional single networks (such as VGG and AlexNet), multi-branch networks show better performance on various vision tasks. For example, the Xception [28] models are one of the most successful multi-branch networks. The usage of multi-branch network structure can increase the receptive field, and thus the convolutional neural network

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obtains more feature information for feature extraction. Therefore, we apply a multi-branch network in our model to learn the multi-scale features extracted from the shallow layer. Specifically, we describe three branches as {branch1, branch2, branch3} (see above Fig.1). Each branch integrates two depthwise separable convolutions, and a Convolutional Block Attention Modules, and extracts features from different scale feature maps. We call this module as DscCBAM module.

Finally, feature fusion is realized through the addition operation. The output feature map of three branches are summed together and input into the next stage. Followed by global the average pooling layer (GAP layer), two depthwise separable convolution layers average the feature map into a scalar value and condense the most core features. For classification, the Softmax function is connected after the GAP layer. The model output is the probability distribution over all categories. Table I shows the content and parameters of the model.

TABLE I
OVERALL ARCHITECTURE OF MODEL

Layer	Input Size	Filter Size	Stride
Input layer	*48*48*	-	-
Conv1	*48*48*64*	*3*3*64*	*1*
Conv2	*48*48*64*	*3*3*64*	*1*
DscCBAM -1	*48*48*64*	*3*3*128*	-
Conv3	*24*24*128*	*3*3*512*	*2*
Conv4	*12*12*512*	*3*3*512*	*2*
DscCBAM-2	*24*24*128*	*3*3*256*	-
Conv5	*12*12*256*	*1*1*512*	*2*
DscCBAM-3	*12*12*128*	*3*3*512*	-
DSC1	*6*6*512*	*3*3*1024*	*1*
DSC2	*6*6*1024*	*3*3*1024*	*1*
GAP	*3*3*1024*	*3*3*	-
FC	*1024*	-	-
Softmax	7/8	classifier	-

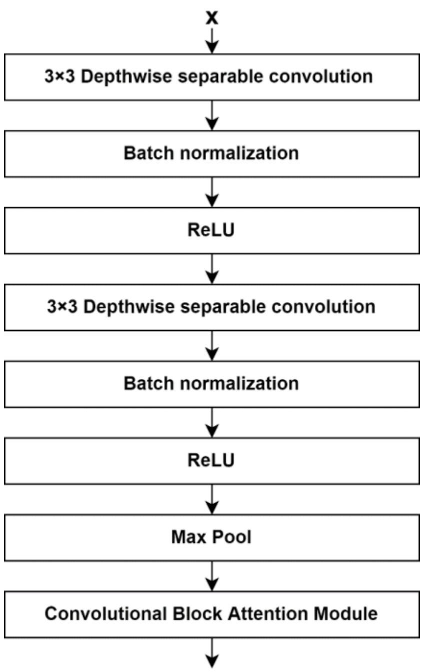

FIGURE 2. Diagram of Dsc-CBAM Module

B. DscCBAM module

In this paper, we proposed DscCBAM module which embedded the CBAM [29] module with depthwise separable convolution, as shown in Fig. 2. The CBAM is added to make the model take care of the important feature in channel and space. Each DscCBAM module composes of 2 depthwise separable convolution layers, Batch normalization, Relu, Max Pooling, and CBAM module.

(1) DEPTHWISE SEPARABLE CONVOLUTION LAYERS

To improve recognition accuracy without increasing model computation and parameters. Inspired by Mobilenet V1, we replace the 3×3 standard convolution with depthwise separable convolution layers and construct a lightweight CNN. Depthwise separable convolution splits a standard convolution into a depthwise convolution and a pointwise convolution with 1×1 convolution kernel [10]. The difference of depthwise separable convolution and standard convolution is shown in Fig. 3.

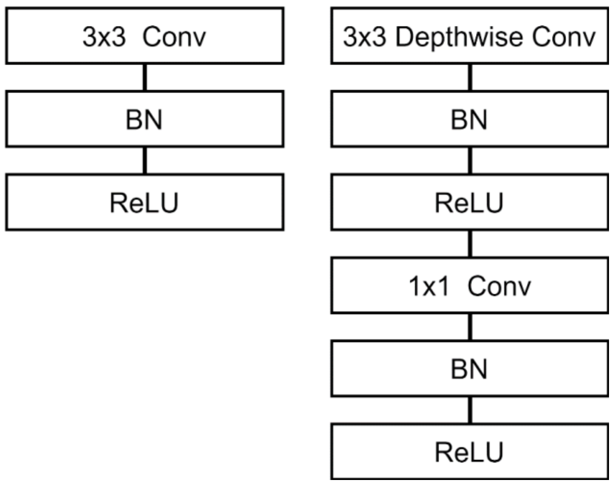

FIGURE 3. The difference of depthwise separable convolution (right) and standard convolution (left)

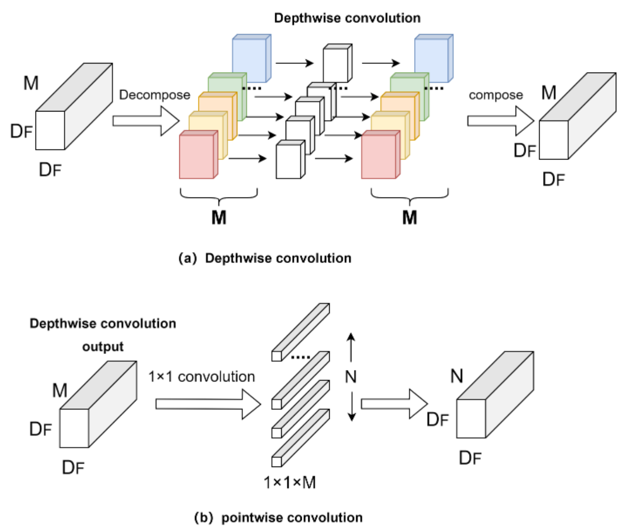

FIGURE 4. (a) Depthwise convolution, (b) pointwise convolution

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Depthwise separable convolution operation consist of two steps. In the first step, the depthwise convolution processes input feature maps separately by different channels with the convolution filter, as depicted in Fig. 4a. Secondly, the output of depthwise convolution is combined linearly with a 1×1 pointwise convolution, as shown in Fig. 4b.

In practice, MobileNet uses 3×3 convolution kernel in depthwise separable convolutions and reduces computation amount to about 1/ 9 of standard convolutions. In our model, depthwise separable convolution is employed to replace standard convolution and reduces parameters and the computation amount. After each depthwise separable convolution layer, there is a ReLu activation function and a Batch Normalization [30]. ReLu is adopted in this paper due to its excellent performance. The calculation of the Relu activation function is shown in Equation (1).

$$f(x) = \begin{cases} 0, x < 0 \\ x, x > 0 \end{cases} \tag{1}$$

The Batch Normalization is added to normalize neuron activations to zero mean and unit variance, which improves training speed and prevents gradients from diminishing and vanishing.

(2) CBAM MODULE

In most image recognition tasks, it is significant to extract essential features from input images. In facial expression recognition, the eyes, nose, and lips are important regions. We can infer emotional states from geometric of these regions. Considering that the different regions of the human face have different degrees of influence on final expression recognition, we should focus on these regions. It has been proven that the attention mechanism is effective for computer vision applications. Attention mechanism is performed to find out the important features which are exploited for the recognition of expression.

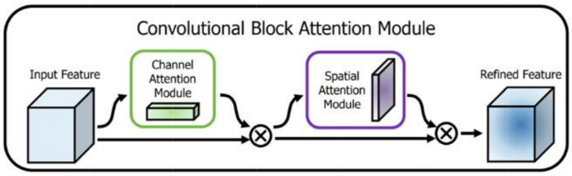

FIGURE 5. The overview of CBAM. The module contains a channel attention module (CAM) and a spatial attention module (SAM) [29]

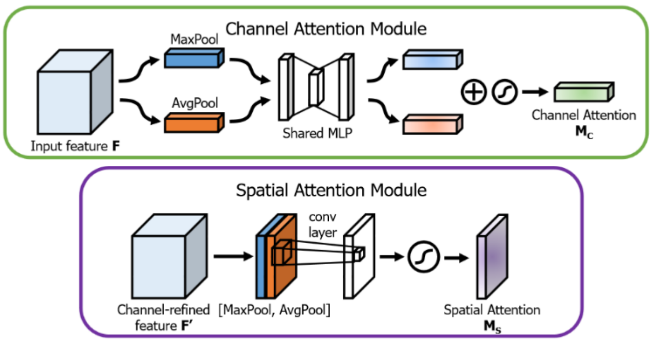

FIGURE 6. Diagram of channel attention module (CAM) and spatial attention module (SAM) [29]

In [29] Sanghyun Woo et al. presented CBAM (Convolutional Block Attention Module). CBAM is based on a lightweight attention mechanism. As Fig. 5 show, CBAM consists of a CAM (channel attention module) and a SAM (spatial attention module). The CBAM attention operation can be represented in Equation (2).

$$\begin{aligned} \mathrm{F}' &= \mathrm{M}_{\mathrm{CAM}}(F) \otimes F, \\ \mathrm{F}'' &= \mathrm{M}_{\mathrm{SAM}}(F') \otimes F' \end{aligned} \tag{2}$$

In Equation (2), F presents the input feature map, $\otimes$ denotes element-wise multiplication, $\mathrm{M}_{\mathrm{CAM}}$ presents the channel attention extraction operation, $\mathrm{M}_{\mathrm{SAM}}$ presents the spatial dimension extraction operation. Fig. 6 demonstrates the process of CAM and SAM.

CAM performs average pooling and max pooling on the feature map to obtain two spatial context feature maps: F_{avg}^{c} and F_{max}^{c}. Then, both two feature maps are sent into a multi-layer perception (MLP) network, and an elementwise summation method is taken to realize the fusion of the output feature maps of MLP. At last, the sum feature is fed to the sigmoid activation, and the output feature M_c is obtained. The channel attention is computed as Equation (3).

$$\begin{aligned} M_c(F) &= \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F)) \\ &= \sigma(W_1\left(W_0(F_{avg}^{c})\right) + \sigma(W_1(W_0(F_{max}^{c})) \end{aligned} \tag{3}$$

In Equation (3), F is the input feature map, σ is the sigmoid function, W_0 and W_1 are the MLP weights.

The SAM learns the spatial information according to the output feature map of the CAM. Firstly, the spatial attention module applies average pooling and max pooling operations and obtains two feature maps: F_{avg}^{s}, F_{max}^{s} . Then, a new feature map is produced by concatenating the two feature maps. Then, a 7×7 convolution layer with a sigmoid function is adopted to yield a spatial attention map M_s. The spatial attention is computed as Equation (4):

$$\begin{aligned} M_s(F) &= \sigma(f^{7\times7}([AvgPool(F)); MaxPool(F)])) \\ &= \sigma\left(f^{7\times7}([F_{avg}^{s}; F_{max}^{s}])\right) \end{aligned} \tag{4}$$

where σ stands for the sigmoid activation function, $f^{7\times7}$ represents a 7×7 convolution operation.

C. FUSION OF BRANCH FEATURES

As shown in Fig. 1, Our model is a neural network containing 3 branches of DscCBAM output. The branch1 adopts two DSC with 128 kernels, one MaxPool2D, one CBAM module, and two DSC with 512 kernels. The branch2 performed four times Dsc layers, two MaxPool2D, one CBAM module, and a DSC with 512 kernels to rescale to the same size. Branch3 passes through six DSC layers, three MaxPool2D, and one CBAM module. The Feature maps of three branches are rescaled to the same size for feature fusion. Multi-branch structure can improve the diversity of feature extraction for hierarchical feature extraction.

Inspired by ResNet, this paper adds a shortcut connection in the middle of the DscCBAM module, and then inputs into the next DscCBAM module. So, we can share parameters with different branches to transmit features efficiently and use less parameters. Meanwhile, the participation of the shortcut connection avoids gradient dissipation or explosion. Finally, the output features of the three branches are combined with element-wise summation. The combined features are sent to two depthwise separable convolution layers, Max pooling layers sequentially.

D. Output and Softmax function

The fully connected layers are the standard component in the classification network. However, some issues are with fully connected layers, e.g., fully connected layers have dense transformation matrices, and too many parameters reduce the speed of training. The last is that it is easy to cause over-fitting. Instead of using the fully connected layer, Lin et al. [31] used the proposed global average pooling and acquired state-of-the-art classification performance on the CIFAR datasets. Global average pooling operation averages the activation map of the last layer into a scalar value and fed the result into the classification layer. A lot of modern architecture reduce parameters and regularize the entire network to prevent over-fitting by adding global average pooling. Therefore, we apply global average pooling processing in our model and perform on the fusion features maps to obtain the feature vector G_{avg}.

After global average pooling layer, this paper adds a fully connected layer as the model's top layer. Because the Softmax function is often taken to predict the probabilities associated with a multinoulli distribution. So, the Softmax function is taken to resolve the multiple classification problems of FER. For a given class i, the output of the softmax can be calculated in Equation (5).

$$P_i = \frac{\exp\,(x_i)}{\sum_{j=1}^{n} \exp\,(x_j)} \qquad (5)$$

By plugging feature maps of the previous layer G_{avg} into the softmax layer, the final facial expression polarity can be obtained, as shown in Equation (6).

$$\mathrm{y} = soft\max\left(w_c G_{avg} + b_c\right) \qquad (6)$$

where w_c is the weight matrix, and b_c is the bias.

Finally, the categorical cross-entropy function is chosen to predict the type of Facial Expression, as shown in Equation (7).

$$L_S = -\sum_{x} \left(y(x) \log z(x)\right) + \left(1 - y(x)\right) \log\left(1 - z(x)\right), \qquad (7)$$

where L_S is the cross-entropy loss function, y represents the dataset's real probability distribution, z represents the probability distribution of the model output.

E. Dropout

Although we use data augmentation to expand the datasets, the number of pictures in some categories is still relatively small. To avoid over-fitting, a dropout strategy was applied in this work. Dropout [32] is an efficient way to make the model more robust. On each training epoch, each hidden neuron is randomly omitted with a probability. This technique reduces the number of neurons that participate in the training step.

IV. EXPERIMENTS

Classification experiments were conducted on FER2013, FERPLUS, CK+, and the experimental results were compared with those of different models. The result will be described in this section.

A. Experimental environment

Experiment was carried out on keras with TensorFlow as the backend. The program and algorithm were designed on Python 3.8.6 and jupyter-notebook tools. The following computer specification has been experimented: Windows 10 operating system, processor Intel Xeon silver 4216, memory 64GB RAM, graphics card NVIDIA RTX3080.

B. DATASETS

The proposed model was trained with three public datasets: Facial Expression Recognition 2013 Dataset (FER2013) [33], FERPLUS [34], and Extended Cohn-Kanade Dataset (CK+) [35]. Sample images of three datasets are shown in Fig. 7.

FIGURE 7. Example Images in FER2013/FERPLUS (top), CK+ (bottom)

1) The FER 2013 is a static facial dataset which provided by ICML2013 Workshop. There are 28,709 images, 3,589 images, and 3,589 images for training, validation, and testing, respectively. The images were divided into seven types: angry, disgust, fear, happy, sad, surprise, neutral. All images are 48 × 48 pixel grayscale images of faces.
2) The FERPLUS dataset is mainly derived from FER2013 dataset. Due to the non-face data and false tags in FER2013, the FERPLUS provide a set of new labels and removes some uncertain images in FER2013. Images

were labeled into 8 emotion types including neutral, happiness, surprise, sadness, anger, disgust, fear, and contempt. Finally, the FERPLUS consists of 35,485 facial grayscale images.

3) The extended Cohn–Kanade(CK+) dataset contains 593 sequences obtained from 123 subjects. The images have 640 × 480 pixel resolutions with gray-scale values. Each image is labeled with one of the seven expressions: anger, contempt, disgust, fear, happy, sad, and surprise. For the CK+ dataset, 5-fold cross validation was performed in the experiment for performance evaluation. All images were shuffled and separated into five equal parts on average. And then rotate among them as training dataset and test dataset. The accuracy of the model was calculated by averaging the five rotating test results.

C. IMAGE PREPROCESSING

The pixel of all images in three datasets are rescaled to the [0,1] interval. Image data augmentation is a powerful technique which used universally in computer vision. Because the small number of CK+ datasets may cause deep learning-based model over-fitted easily. The data-augmentation techniques, including horizontal flip, vertical flip, random zoom transformation, random rotation transformation, and Shear rage, are used to generate the augmented dataset and expand the number of samples.

D. HYPER-PARAMETERS SETTING

In the network model training phase, the optimization method used Adam, and the initial learning rate is set to 0.001. During the training process, if the validation datasets loss does not reduce the specified value, the learning rate decay is applied to the current value. The iteration batch size is 64. The maximum epoch number is 300, and the average accuracy of the last 100 epochs from the 200th to the 300th epoch is defined as the final convergence accuracy (FCA).

V. RESULTS AND DISCUSSION

To evaluate the proposed model, we consider some influential CNNs (et al. AlexNet, VGG16, GoogleNet, ResNet50, DenseNet, Xception, and EfficientNet) and other excellent models for the comparative experiments.

Facial expression recognition is a multi-classification task. The performance of the models is evaluated in terms of accuracy, as defined in Equation (8):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \qquad (8)$$

Where TP indicates the number of images that the classifier identifies correctly; TN indicates the number of samples predicted correctly as negative; FP indicates the number of images which are miss-classified as positive classes; FN indicates the number of images that are miss-classified as negative class.

A. FERPLUS data set Experiment.

Table 2 shows the comparison results of our model with other methods. Our method achieves 84.633%. It is 2.633%, 4.733%, and 0.343% higher than the references [21,24,25]. To evaluate the performance, the proposed model's confusion matrix on the FERPLUS dataset is shown in Fig. 8. The accuracy of our method in recognizing anger, disgust, fear, happiness, sadness, surprise, neutral, and contempt expressions is 80%, 43%, 46%, 90%, 58%, 85%, 89%, and 30% respectively. Meanwhile, both "disgust" and "contempt" are easier to be misclassified with low accuracies. This could be explained by lack of training samples of two categories. Fig. 9 illustrated the accuracy and Loss of both validation(public) and test(private) on FERPLUS data set. It can be observed that a stable growth accuracy trend and loss decline trend before 50 epochs.

TABLE II
COMPARISON OF DIFFERENT METHODS ON THE FERPLUS DATASET

Model Name	Accuracy
VGG16	77%
Zheng Lian et al [21]	82%
ResNet50 (from scratch) [24]	54.43%
TFE-joint learning. [25]	84.29%
ResNet50 (transfer learning) [24]	79.9%
DenseNet	84.06%
Xception	83.54%
EfficientNet (B4)	81.51%
The proposed algorithm	**84.633%**

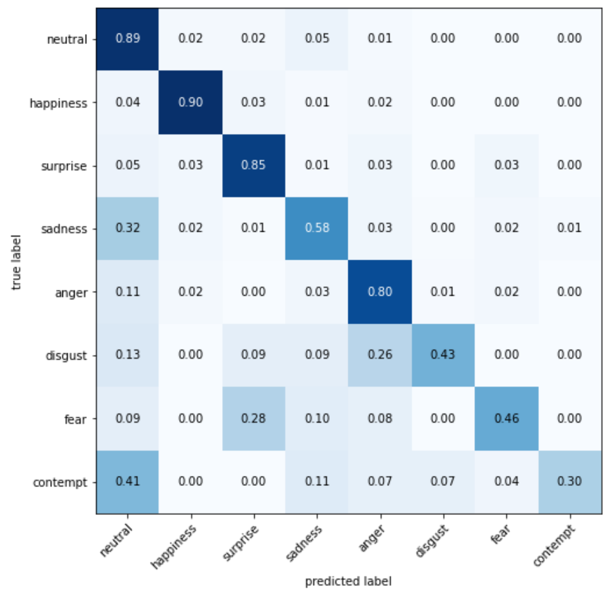

FIGURE 8. **Confusion matrices on FERPLUS dataset**

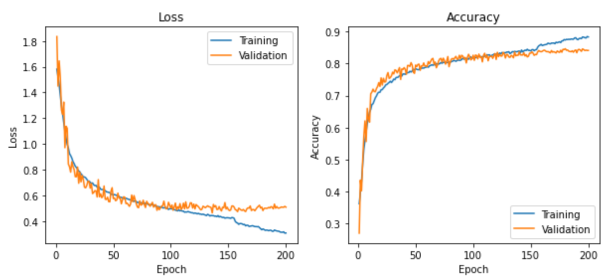

FIGURE 9. **The Accuracy-Loss of our model on FERPLUS dataset**

B. FER2013 data set Experiment.

To validate our model's performance, the experiments are implemented on FER2013 too. The details of the confusion matrix of the result for a seven-classes classifier is given in Fig. 10. The Accuracy-Loss and the confusion matrixes are shown in Fig. 11. We design comparison experiments of the proposed model with VGG16, GoogleNet AlexNet, Bag of Visual Words, and CNN-SIFT Aggregator et al., and the comparison results are listed in Table 3. The recognition accuracy of this method is 69.49%. It can be seen from Table 3 that the expression classification and recognition accuracy of all methods are below 74% on the FER2013 dataset. This is due to the non-face datas and false tags in FER2013.

TABLE III
COMPARISON OF DIFFERENT METHODS ON THE FER2013 DATASET

Model Name	Accuracy
VGG16	61.55%
GoogleNet [14]	65.2%
AlexNet [16]	61.1%
Mollahosseini et al. [16]	66.4%
CNN-SIFT Aggregator [17]	73.4%
Bag of Visual Words [18]	67.484%
NING ZHOU et al.[20]	67%
Shervin Minaee et al. [24]	70.2%
The proposed algorithm	**69.49%**

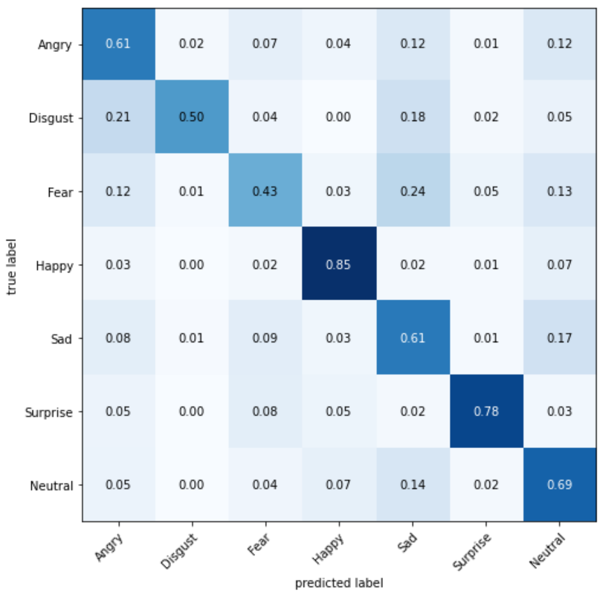

FIGURE 10. Confusion matrices on FER2013 dataset

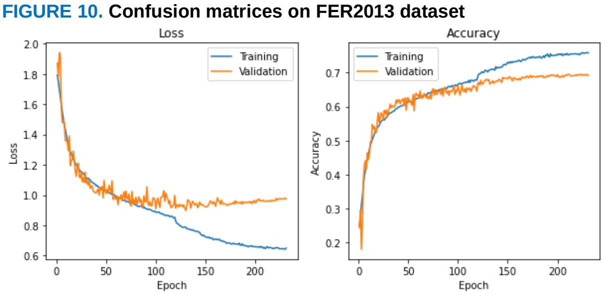

FIGURE 11. The Accuracy-Loss of our model on FER2013 dataset

C. CK+ data set Experiment.

Because CK+ data set does not provide specified training, validation, and test sets, we used K-fold cross-validation (typically K=5) to reliably evaluate our model. The experiments are repeated five times, and the final accuracy is the average result. The confusion matrix on the CK+ data set is shown in Fig. 12. We compared the model with ResNet50, VGGNet, AletNet, CNN-SIFT Aggregator, et al. Table 4 indicates that the accuracy rate is better than other methods. As shown in Fig. 12, The CK+ dataset sample is relatively balanced in the accuracy of each class. And We were able to achieve nearly 100% accuracy. Compared with other methods, our method performs better.

TABLE IV
COMPARISON OF DIFFERENT METHODS ON THE CK+ DATASET

Model Name	Accuracy
Yanpeng Liu et al. [11]	98.3%
VGGNET [15]	83.57%
AlexNet [16]	92.2%
Mollahosseini et al. [16]	93.2%
CNN-SIFT Aggregator [17]	99.1±0.07%
Shervin Minaee et al. [23]	98%
Mengyi Liu et al. [36]	95.78%
Cornejo et al. [37]	99.78%
The proposed algorithm	**99.39%**

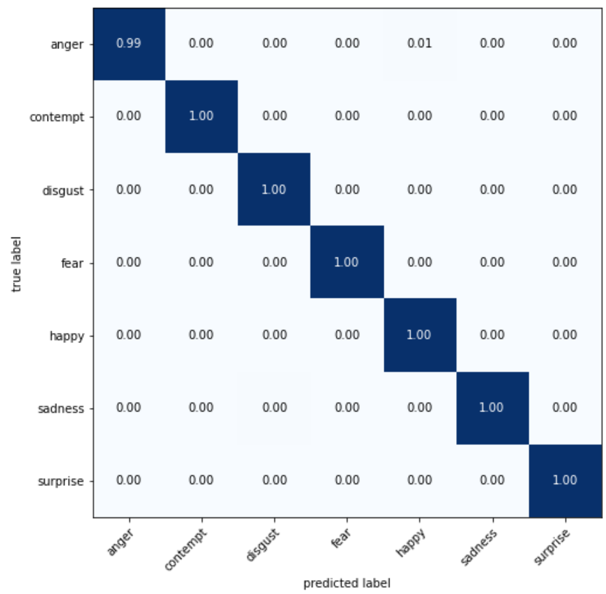

FIGURE 12. Confusion matrices on CK+ dataset

D. DISCUSSION

The accuracy and loss curves of the proposed model on the FERPLUS and FER2013 datasets are shown in Fig. 9 and Fig. 11. We can intuitively see that the accuracy and loss curves tend to balance after 50 periods and fluctuates in a small range. The results in Tables II, III, and IV demonstrated that our proposed model can achieve a higher recognition rate than some other methods. The reason is that our model explicitly takes the advantage of multi-branch architecture and attention mechanisms, and hence improves recognition accuracy.

Fig. 8, Fig. 10, and Fig. 12 show the confusion matrix that includes the accuracies per class. According to the confusion matrices, the proposed model has high recognition accuracy for neutral, happiness, surprise, and anger classes. However, we can observe that the proposed model gets poor recognition accuracy for the rest classes. The reason is that some of them have similar texture features around the key region. This makes it hard for the model to identify. We will continue to explore new methods to solve this problem and achieve better performance.

VI. CONCLUSION

This paper develops a multi-branch attention CNN for facial expression recognition, which combines the Convolutional Block Attention Module, Depthwise separable convolution, and three multi-branch structure approaches. The Convolutional Block Attention module is exploited to enable a better extracting of features from facial expression images, and depthwise separable convolution is added to reduce the model parameters. We compared the accuracy of developed using FERPLUS, FER2013, and CK+ datasets with the classic classification models. The recognition accuracy is 84.633%, 69.49%, and 99.39% on the FERPLUS, FER2013, and CK+, respectively. It has been found that the model proposed in this paper provides better expression recognition performance and generalization capability, promising a further improvement for real video data.

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